Applying Machine Learning Advances to Data Visualization: A Survey on ML4VIS

Qianwen Wang, Zhutian Chen, Yong Wang, and Huamin Qu

Abstract—Inspired by the great success of machine learning (ML), researchers have applied ML techniques to visualizations to achieve a better design, development, and evaluation of visualizations. This branch of studies, known as ML4VIS, is gaining increasing research attention in recent years. To successfully adapt ML techniques for visualizations, a structured understanding of the integration of ML4VIS is needed. In this paper, we systematically survey 85 ML4VIS studies, aiming to answer two motivating questions: *"what visualization processes can be assisted by ML?"* and *"how ML techniques can be used to solve visualization problems?"* This survey reveals six main processes where the employment of ML techniques can benefit visualizations: *VIS-driven Data Processing, Data Presentation, Insight Communication, Style Imitation, VIS Interaction, VIS Perception*. The six processes are related to existing visualization theoretical models in an ML4VIS pipeline, aiming to illuminate the role of ML-assisted visualization in general visualizations. Meanwhile, the six processes are mapped into main learning tasks in ML to align the capabilities of ML with the needs in visualization. Current practices and future opportunities of ML4VIS are discussed in the context of the ML4VIS pipeline and the ML-VIS mapping. While more studies are still needed in the area of ML4VIS, we hope this paper can provide a stepping-stone for future exploration. A web-based interactive browser of this survey is available at https://ml4vis.github.io.

Index Terms—ML4VIS, Machine Learning, Data Visualization, Survey.

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1 Introduction

Data visualization (VIS), which maps abstract data into visual presentations (*e.g.*, position, color), is a powerful approach to uncover hidden insights and communicate compelling stories in data. However, achieving an effective visualization is typically challenging, requiring a large amount of human effort and having a high reliance on expertise, such as graphic design, user experience design, and data analysis.

Machine Learning (ML), on the other hand, provides a practical opportunity to relieve the reliance on experts in visualization. By automatically learning knowledge from data, ML enables task completion without explicit instructions from humans. A variety of visualization-related activities, including the creation, interaction, and evaluation of visualizations, can benefit from the application of ML techniques. We refer to studies that *apply ML techniques to assist visualization-related activities using the knowledge extracted from data* as **ML4VIS** in this paper. ML4VIS studies can be traced back to 1986 [1] and have been reignited by the recent advances in ML. A series of ML4VIS studies are emerging, covering a wide range of visualization problems (*e.g.*, graph layout [2], visualization assessment [3]) and ML techniques (*e.g.*, variational auto-encoders [4], Mask-RCNN [5]).

While ML4VIS generates a stream of new opportunities, it also poses a series of challenges. First, it is not always clear, among various problems related to visualization, which one can be significantly improved by existing ML techniques and which one still requires a high level of human intervention. Roughly applying ML techniques to unsuitable visualization problems may only impose the drawbacks of ML (*e.g.*, uncertainty, inexplainability) and threaten the validity of a visualization without bringing any benefit. Second, given a visualization-related task, selecting a proper ML technique and employing necessary adaptation are crucial yet challenging. There are a plethora of ML techniques, most of which are exclusively suitable for a certain type of problems. For example, to automatically interpret infographics, ML techniques developed for natural-image-understanding are more helpful and relevant than ML techniques developed for speech recognition. Therefore, the success of ML4VIS hinges on a better understanding of both the visualization and the ML, as well as the integration between the two.

Researchers have contributed a series of frameworks and surveys about the integration between visualization and ML, but they mainly investigate how visualization can assist ML in data analysis [6], [7], [8] rather than how ML can be used to solve visualization-related problems. For example, in a state-of-the-art report, Endert et al. [8] depicted a closer integration between visualization and ML techniques through interactive ML. However, the goal of this integration is still to facilitate data analysis in different application domains (*e.g.*, text analytics, multimedia analytics).

Some recent surveys have reviewed studies on automatic visualization creation [9], [10], but these surveys consider both ML-based and non-ML-based methods. More importantly, they only focus one specific visualization-related problem, *i.e.*, visualization creation. The opportunities for applying ML techniques to various visualization-related problems (*e.g.*, visualization creation, visualization assessment) have not been fully discussed in previous studies. To the best of our knowledge, this paper is the first published survey on ML4VIS.

In this work, we systematically reviewed the literature in related

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fields including data visualization, human-computer interaction (HCI), ML, and data mining to investigate how ML techniques are employed and adapted for visualization. Build upon existing visualization theoretical models, we proposed in an ML4VIS pipeline, which reveals six main processes that can benefit from employing ML techniques. The six visualization processes (*i.e.*, *VIS-driven Data Processing, Data Presentation, Insight Communication, Style Imitation, VIS Interaction, VIS Perception*) are mapped into the main learning tasks in ML to bridge the needs in visualization with the capabilities of ML, we map the six processes to main learning tasks in ML. We discuss the current practices and future opportunities of ML4VIS in the context of the ML4VIS pipeline and the ML-VIS mapping. Specifically, our contributions are:

- A codification of 85 ML4VIS studies into **six processes** in visualization that can benefit from the employment of ML techniques (Section 4);
- **An ML4VIS pipeline** that integrates the six processes with existing visualization model (Section 5) and an **ML-VIS mapping** between the six processes and an ML taxonomy (Section 6).
- A set of **research challenges and opportunities** derived from our analysis (Section 7).

We hope this paper can provide a stepping stone for further exploration in the area of ML4VIS. An interactive browser of this survey is available at https://ml4vis.github.io.

2 Related Work

This paper mainly relates to two streams of literature: surveys that aim to guide ML-assisted design, theoretical models that summarize the process in visualization.

2.1 Guiding ML-assisted Design

As various ML capabilities (*e.g.*, recommendation, interaction prediction) are integrated into user interfaces, many surveys and reviews have been conducted to guide the application of ML techniques in the design of user interfaces [34], [35], [36].

These surveys on general ML-assisted designs have contributed various guidelines and taxonomies that also inform the ML4VIS studies. Efforts to review these guidelines can be traced back to 1999, when Horvitz [36] reviewed and outlined 12 guidelines for coupling automated services with direct user manipulation. These guidelines were later extended by Amershi et al. [35], who distilled 18 design guidelines for human-AI interaction from over 150 AI-related design recommendations. The majority of these guidelines, such as *"learn from user behaviors"*, *"convey the consequence of user interactions"*, can also be applied to the design of ML-assisted interactive visualizations. Apart from design guidelines, taxonomies can also provide a better understanding of ML techniques through a design perspective. Yang et al. [34] analyzed more than 2000 HCI publications using topic modeling and summarized four channels through which ML advances can provide value to users: self-understanding, contextual awareness, optimization, and utility-capability. The four channels can also help people envision new ways of ML techniques to improve visualization. While these surveys shed some light on the study of ML4VIS, visualizations have certain unique characteristics that require specified investigations. For example, a precise understanding of and an efficient interaction with data are required in visualization, but they are rarely discussed in the general design of user interfaces.

To guide ML-assisted visualization designs, Saket et al. [37] compared and categorized existing approaches for learning visualization design principles based on the learning method and the input features. The authors then described a research agenda for deriving visualization design principles using ML techniques. Even though this study provides a helpful and insightful agenda, it focuses on the creation of visualization and makes little investigation about user interaction and perception, which are important in interactive visualizations. In this study, we present a survey on ML4VIS that covers various visualization processes.

2.2 Theoretical Models in Visualization

Theoretical models (*e.g.*, workflows, pipelines) have become increasingly important in designing, developing, and evaluating visualizations. A variety of theoretical models have been proposed to depict different processes in visualization [7], [38], [39]. For example, Card [40] presented a visualization workflow to describe the process of creating and interacting with visualizations. Munzner [38] proposed a nested model for the design and validation of visualization.

With the growing usage of ML in visualization, more and more studies have included ML techniques as a component in the theoretical models of visualization. Keim et al. [7] presented the first effort to form a general VA pipeline, where ML is included as a data processing module. Follow-up studies proposed theoretical models tailored for different analysis scenarios and ML techniques [41], [42], [43]. For example, El-Assady et al. [43] extended the general VA pipeline to encompass a new visual analytics paradigm, *i.e.*, speculative execution. Sacha et al. [41] introduced a process model to describe the visual interaction with dimension reduction, a widely-used ML technique. Endert et al. [8] reviewed theoretical models that embed ML techniques into visual analytics and listed opportunities for future research. In these studies, ML is integrated with visualization techniques to facilitate the analysis of complex data. The capabilities of ML in enabling more effective visualizations still remain unclear.

Recently, there is a growing research trend on modeling the role of visualization in the design, development, and evaluation of ML. For example, a Dagstuhl Seminar [6] identified a series of research opportunities in combining ML with visualization. Sacha et al. [44] reinterpreted the traditional VA pipeline and proposed a VIS4ML ontology to encompass ML development workflows into VA workflows. Spinner et al. [45] presented a VA framework for interactive and explainable ML. A research gap exists in summarizing how ML can contribute to visualization.

Contrary to previous theoretical models, we investigated how ML techniques can be used to benefit the creation, evaluation, and interaction of visualizations. We presented an ML4VIS pipeline based on our survey of 85 papers. While more studies are still needed in the field of ML4VIS, we hope this pipeline can help map out the current landscape and inform future opportunities.

3 Survey Methodology

To provide an understanding of the current ML4VIS studies, we conducted an analysis of 85 related papers in the field of VIS, HCI, ML, and data mining & management. Each paper is coded based on the **VIS process** and the **ML task**. Even though automatic analysis methods (*e.g.*, topic modeling) are popular among recent survey papers [34], [41], [46], automatic methods usually only provide a high-level understanding of a large number of papers (*e.g.*, more than 2000 [34]). Manual analysis is still needed to extract details for

TABLE 1
Relevant venues

VIS (visualization)	IEEE VIS (InfoVis, VAST, SciVis), EuroVis, PacificVis, TVCG
HCI (human-computer interaction)	CHI, UIST, IUI
DMM (data mining and management)	KDD, SIGMOD, ICDE, VLDB
ML (machine learning)	CVPR, ECCV, ICML, NeurIPS, ICCV, AAAI, IJCAI, ICLR

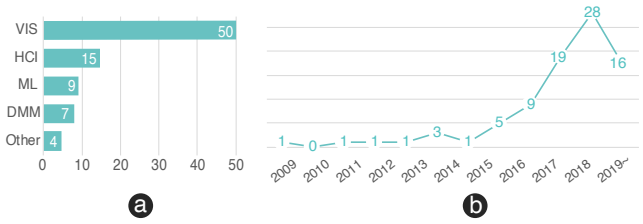

Fig. 1. (a) The number of ML4VIS studies published in different fields: VIS, HCI, DMM (data mining and management), ML;(b) The number of ML4VIS studies over time.

a deep analysis [41], [46]. ML4VIS, as a new emerging topic, has a relatively small body of literature that can be manually analyzed. Therefore, to ensure a thorough analysis, we conducted a manual analysis for the collected papers.

3.1 Survey Scope

We began our analysis by assembling a corpus of ML4VIS papers. We collected all research articles published between 2016 Jan to 2020 Oct in main journals and conferences in the field of VIS, HCI, ML, and data mining & management, as shown in Table 1. This time range is based on two main considerations: 1) an increasing number of research papers on applying ML techniques to visualization appear in the latest five years, 2) given that ML techniques are progressing rapidly, the latest research is more important for informing readers of the latest trend in ML4VIS. We mainly checked the title and abstract of each paper. During this process, we paid special attention to a set of ML-related keywords (*e.g.*, *"learning"*, *"machine"*, *"training"*, *"AI"*, *"automation"*, *"CNN"*, *"LSTM"*) and visualization-related keywords (*e.g.*, *"visualization"*, *"infographic"*, *"diagrams"*, *"charts"*). After such a round of paper selection and filtering, we obtained 259 papers.

We further carefully checked the 259 papers using the following criteria. First, we validated the term **ML**: the paper should use ML techniques to extract knowledge from training data. For example, Voyager [47] was excluded from the corpus because it recommends charts using rules defined by human experts rather than the knowledge learned by ML models. Second, we validated the term **VIS**: the paper should solve visualization-related problems. For example, Liu et al. [48] used ML techniques to learn the design semantics of mobile user interfaces. Since mobile user interfaces have little visualization components, this study was removed from the corpus. Third, we validated the term **4 (for)**: ML should be used for visualization-related problems, such as the creation, interaction, and evaluation of visualizations. This

TABLE 2
All surveyed papers and their codes.

ID	paper	venue		Data Processing	VIS-driven	Present Data	Communicate Insight	Imitate Style	VIS Perception	VIS Interaction	Dimension Reduction	Clustering	Generative	Classification	Regression	Semi-supervised	Reinforcement
1	Gotz and Wen [86]	IUI	2009		■						■						
2	Savva et al. [107]	UIST	2011							■					■		
3	Key et al. [11]	SIGMOD	2012		■										■		
4	Steichen et al. [84]	IUI	2013								■				■		
5	Brown et al. [62]	TVCG	2014								■				■		
6	Lalle et al. [83]	IUI	2014								■					■	
7	Toker et al. [12]	IUI	2014								■				■		
8	Sedlmair and Aupetit [13]	CGF	2015							■					■		
9	Mutlu et al. [14]	TiiS	2016		■											■	
10	Aupetit and Sedlmair [95]	PVis	2016							■					■		
11	Siegel et al. [102]	ECCV	2016							■					■		
12	Kembhavi et al. [92]	ECCV	2016							■					■		
13	Al-Zaidy et al. [15]	AAAI	2016							■					■		
14	Poci et al. [88]	VIS	2017							■					■		
15	Kwon et al. [74]	VIS	2017					■								■	
16	Bylinskii et al. [64]	UIST	2017							■						■	
17	Saha et al. [117]	IJCAI	2017	■									■				
18	Kruiger et al. [16]	EuroVis	2017		■								■				
19	Poco and Heer [89]	EuroVis	2017							■					■		
20	Jung et al. [99]	CHI	2017							■					■		
21	Bylinskii et al. [100]	arxiv	2017							■					■	■	
22	Al-Zaidy and Giles [17]	AAAI	2017							■					■		
23	Siddiqui et al. [61]	VLDB	2018								■				■		
24	Gramazio et al. [85]	VIS	2018								■				■		
25	Moritz et al. [18]	VIS	2018		■										■	■	
26	Berger et al. [68]	VIS	2018		■									■			
27	Wang et al. [53]	VIS	2018	■									■				
28	Haehn et al. [19]	VIS	2018							■						■	
29	Luo et al. [57]	SIGMOD	2018				■								■	■	
30	Milo and Somech [80]	KDD	2018								■			■			
31	Zhou et al. [20]	IJCAI	2018								■					■	
32	Kahou et al. [101]	ICLR	2018							■					■		
33	Luo et al. [65]	ICDE	2018		■										■		
34	[Fan and Hauser [79]	EuroVis	2018								■				■		
35	Chegini et al. [96]	EuroVis	2018							■						■	
36	Kafle et al. [63]	CVPR	2018							■					■	■	
37	Kim et al. [106]	CVPR	2018							■					■		
38	Battle et al. [108]	CHI	2018							■					■		
39	Dibia and Demiralp [54]	CGA	2018		■										■		
40	Haleem et al. [94]	CGA	2018							■						■	
41	Madan et al. [103]	arxiv	2018							■					■	■	
42	Yu and Silva [82]	VIS	2019								■					■	
43	He et al. [69]	VIS	2019		■									■			
44	Chen et al. [59]	VIS	2019								■				■		
45	Han and Wang [67]	VIS	2019		■									■			
46	Chen et al. [55]	VIS	2019		■									■			
47	Kwon and Ma [75]	VIS	2019					■						■			
48	Wang et al. [2]	VIS	2019					■								■	
49	Han et al. [120]	VIS	2019								■		■	■			
50	Wall et al. [111]	VIS	2019								■			■			
51	Fujiwara et al. [118]	VIS	2019	■									■				
52	Fu et al. [3]	VIS	2019							■			■			■	
53	Porter et al. [21]	VIS	2019								■		■				
54	Jo and Seo [119]	VIS	2019							■			■			■	
55	Ma et al. [93]	VIS	2019							■					■		
56	Wang et al. [73]	VIS	2019			■									■		
57	Cui et al. [56]	VIS	2019			■									■		
58	Chen et al. [5]	VIS	2019							■					■		
59	Wang et al. [22]	VIS	2019		■											■	
60	Smart et al. [58]	VIS	2019					■					■				
61	Huang et al. [104]	VIS	2019								■			■			
62	Hong et al. [23]	PacificVis	2019		■									■			
63	Fan and Hauser [122]	EuroVis	2019								■		■				
64	Ottley et al. [60]	EuroVis	2019								■			■			
65	Abbas et al. [121]	EuroVis	2019							■			■	■			
66	Kassel and Rohs [24]	EuroVis	2019		■					■							■
67	Hu et al. [66]	CHI	2019		■										■		
68	Fan and Hauser [25]	CGA	2019								■				■		
69	Kafle et al. [26]	arxiv	2019							■					■		
70	Mohammed [27]	VLDB	2020								■			■			
71	Zhang et al. [90]	VIS	2020							■				■		■	
72	Wu et al. [77]	VIS	2020					■									■
73	Tang et al. [76]	VIS	2020					■									■
74	Qian et al. [28]	VIS	2020			■									■		
75	Wang et al. [29]	VIS	2020	■											■		
76	Fosco et al. [112]	UIST	2020							■						■	
77	Giovannangeli et al. [139]	PacificVis	2020							■						■	
78	Liu et al. [105]	PacificVis	2020							■					■	■	
79	Luo et al. [52]	ICDE	2020	■											■		■
80	Lekschas et al. [113]	EuroVis	2020								■		■		■		■
81	Zhao et al. [30]	CHI	2020					■					■				
82	Lai et al. [31]	CHI	2020							■					■	■	
83	Kim et al. [32]	CHI	2020							■					■	■	
84	Lu et al. [33]	CHI	2020							■					■	■	
85	Zhou et al. [109]	arxiv	2020		■												■

criterion distinguishes ML4VIS from previous VA studies that use ML to analyze data. For example, NNVA [49] employs ML techniques to analyze the parameter sensitivity of yeast cell polarization simulations. In this study, visualizations are merely used to represent the sensitivity analysis that are generated by ML. Therefore, NNVA is not an ML4VIS study and was eliminated from the corpus. Meanwhile, since this survey aims to understand the current practices of employing ML techniques for visualizations, we focus on paper that contribute novel techniques and applications. Theory and evaluation papers were excluded. Papers that used well-established methods are also excluded. For example, a large number of papers that used t-SNE to visualize high-dimensional data are not included. We further went through the related work of these papers and collected the referenced papers that satisfy the aforementioned three criteria. To focus on the latest development in the field of ML4VIS, we did not included referenced papers that were published before 2000. Our final corpus included 85 relevant papers. A summary of the survey statistics is shown in Fig. 1.

3.2 Paper Coding

The paper analysis consists of three phases. In the first phase, we extracted a brief description for each paper, including the targeted visualization problems, the employed ML models, and the collected training data. In the second phase, we coded the collected papers from a visualization perspective, aiming to answer **"what visualization processes can be automated by ML?"**. We referred to the visualization processes in existing visualization pipelines [39], [40] and modified them accordingly to better fit the context of ML4VIS. Each of the three authors independently coded the content of 60 papers to ensure that each paper was coded by at least two authors. The three authors then discussed the codes with all authors. We iteratively adapted, split, and refined the codes multiple times during discussions and finally arrived at six main visualization processes. More details of the six visualization processes are explained in Section 4. Meanwhile, the six processes form an ML4VIS pipeline, whose differences from existing visualization pipelines are discussed in Section 5. In the third phase, we sought to characterize **"how ML can be used to solve visualization problems?"** As new ML models constantly emerge and one ML model is usually applied to solve various problems, it is not practical to code ML4VIS studies based on the used ML models. Therefore, we answered this question by coding the types of learning tasks. As with phase two, each of the three authors coded 60 papers to ensure each paper was at least coded by two authors. We employed a bottom-up approach and referred to the learning tasks in [50], [51]. The learning tasks are summarized and discussed in Section 6. The codes of all collected papers are summarized in Table 2. Most of the ML4VIS papers used one ML technique to solve one major visualization problem. But it is also possible that an ML4VIS paper involves more than one learning task and one visualization process, so a few papers are labeled with more than one ML or VIS code.

4 Benefiting from ML: Six Visualization Processes

In this section, we introduce six visualization processes emerged from our literature review that are benefiting from ML techniques. As shown in Table 3, we first explain the transition from inputs to outputs at each visualization process, and then discuss the main purposes of employing ML techniques at each process with representative examples.

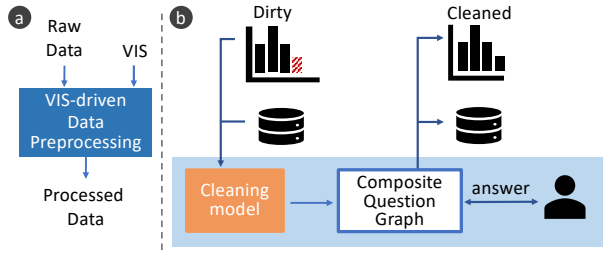

Fig. 2. (a) An illustration of the *VIS-driven Data Processing*; (b) An example of employing ML in *VIS-driven Data Processing*: Luo et al. [52]. An ML model (orange box) is trained to detect data errors and propose cleaning options.

4.1 VIS-driven Data Processing

Process Description: In *VIS-driven Data Processing* (Fig. 2(a)), raw data is transformed into a format that better suits the following visualization processes. Contrary to the general-purpose data processing, *VIS-driven Data Processing* refers to a process that is tightly related to and exclusively designed for the context of visualization, such as enabling efficient visualization creation and enhancing human perception.

Purposes of Employing ML: In *VIS-driven Data Processing*, ML techniques demonstrate the ability to make raw data better satisfy certain visualization-related purposes, including enabling efficient visualization creation and enhancing human perception.

▸ *Enabling efficient visualization creation.* By narrowing down the purpose of data processing to the creation of specific visualizations, the employment of ML techniques can improve the efficiency of data processing. For example, data cleaning, an inevitable step in data processing, is often treated as an independent step and completed before the visualization creation. Instead, Luo et al. [52] related the data cleaning processes with visualization creations and used ML techniques to help users interactively clean the data after the visualization creation. Fig. 2(b) illustrates an overview of [52]. Users first specify a visualization and create it using the uncleaned data. A trained model then detects errors and generates cleaning options for the data underlying the visualization. These cleaning options are provided to users in the form of yes/no questions. Based on users' answers, the underlying data is cleaned and the visualization is updated. This progressive data cleaning enables a more flexible and efficient creation of visualization, since data doesn't need to be totally cleaned before creating visualizations.

▸ *Enhancing human perception.* Meanwhile, ML techniques also demonstrate the ability to enhance user perception of the created visualizations in *VIS-driven Data Processing*. For example, dimension reduction (DR) is a widely used data processing method for visualizing high dimensional data. Previous studies often directly employ the general DR methods proposed in ML field without considering the needs in visualizations. Wang et al. [53] proposed a supervised DR method that takes the perceptual capabilities of humans into account. The modified DR method mimics human perception in cluster separation and successfully maximizes the visually perceived cluster separation in 2D projections.

4.2 Data Presentation

Process Description: In *Data Presentation* (Fig. 3(a)), the processed data is transformed into appropriate visual representations in order to help people better understand and analyze the data. This

TABLE 3
Six main visualization processes that benefit from employing ML techniques.

	Process	Description	Main Purposes of Employing ML	Representative Example
DATA	VIS-driven Data Processing	raw data is transformed into a format that better suits the following visualization processes	improve the efficiency of data processing	ML cleans data for a specified visualization by asking questions to users [52]
			enhance perception of generated visualizations	ML transforms high dimension data to 2D and maximizes the human-perceived separation among classes [53]
VIS	Data Presentation	visualizations are created to present the already processed data	generate suitable visualizations	ML generates Vega-Lite visualization specifications for a given dataset [54]
			improve the efficiency of visualization creation	ML synthesizes intermediate density map images between given density maps without storing and visualizing data of all time steps [55]
	Insight Communication	visualizations are created to communicate the given insights	interpret insights	ML recognizes entities from the user-provided text insights [56]
			generate suitable visualizations	ML predicts suitable visualization specifications for communicating given insights [57]
	Style Imitation	visualizations are created with similar visual properties of the given examples	imitate the color selection	ML generates color ramps by imitating designers' practices in choosing colors [58]
			imitate the layouts	ML generates graph drawings with similar layout styles of the given examples [2]
USER	VIS Interaction	users interact with a visualization and transformed it into a new stage through user interactions	refine the result of the current interaction	ML refines the selected points of a lasso selection in 3D point clouds [59]
			predict next-step interactions	ML predicts a user's next click in a scatterplot [60]
			understand natural interactions	ML interprets the given text descriptions and extracts the corresponding visual patterns [61]
			learn about users	ML predicts a user's task performance and personality based on mouse interaction [62]
	VIS Perception	users read visualizations and obtain useful information	extract content	ML extracts an extensible visualization template from a timeline infographics [5]
			interpret content	ML answers questions about a given bar chart visualization [63]
			model perception	ML predicts the distribution of users' attention on an infographic [64]

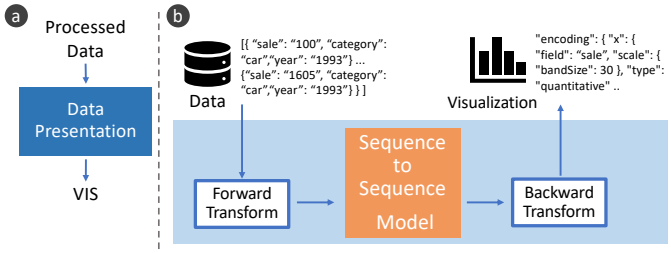

Fig. 3. (a) An illustration of the *Data Presentation* process. (b) A representative example of employing ML in *Data Presentation*: Data2Vis [54]. An ML model (orange box) is trained to map JSON-encoded datasets into Vega-lite visualization specifications.

process is achieved through a systematic mapping from data values to graphic marks. This mapping is usually manually specified using code or authoring tools, which results in steep learning curves and makes data visualization inaccessible to general users.

Purposes of Employing ML: In the collected ML4VIS studies, ML techniques mainly facilitate *Data Presentation* by recommending suitable visual representations and by improving the efficiency of visualization creation.

▸ *Recommending visualizations.* A set of recent research has shown that ML can be used to automatically recommend suitable visual representations. For example, DeepEye [65] combines supervised ML techniques with expert rules to automatically recommend good visualizations for users. Given a dataset, all the possible visualizations are enumerated and classified as "good" or "bad" by a binary decision tree classifier. All the "good" visualizations are then ranked and provided to users. Meanwhile, the great success of deep learning in various domains promotes its application in *Data Presentation*. VizML [66] collects a large visualization dataset and trains a fully-connected neural network to predict the top five design choices when creating visualizations for a specific dataset. Data2Vis [54] formalizes the process of *Data Presentation* as a sequence to sequence translation from the original data to the visualization specifications. As illustrated in Fig. 3(b), a sequence-to-sequence neural network learns the mapping between JSON-encoded datasets (source sequence) and Vega-lite visualization specifications (target sequence). In these studies, ML automatically learns design choices from the training dataset and directly translates data to suitable visualizations. As a result, the employment of ML can effectively reduce the manual efforts in selecting data attributes, designing visual representations, and specifying the mapping from data values to graphic marks.

▸ *Improving the efficiency of visualization creation.* Another line of research focuses on improving the efficiency of creating visualizations. One direction is to improve the efficiency of creating dynamic visualizations. For instance, creating dynamic density maps usually leads to high computational cost and memory demand since a large amount of data needs to be recorded

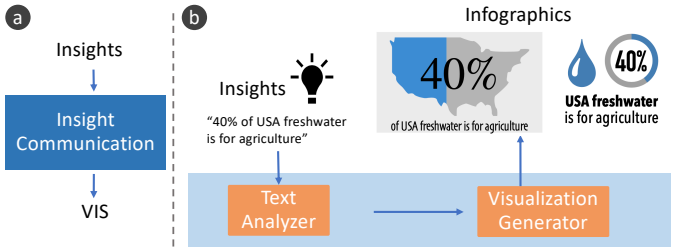

Fig. 4. (a) An illustration of the process of *Insight Communication*. (b) One representative example of ML4VIS in *Insight Communication*: Text-to-Viz [56]. A text insight is first analyzed by a text analyzer and then be transformed into expressive infographics by a visualization generator.

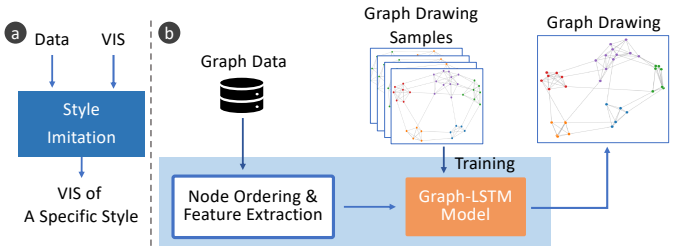

Fig. 5. (a) An illustration of the process of *Style Imitation*. (b) A representative example of ML4VIS in *Style Imitation*: DeepDrawing [2]. A Graph-LSTM model (orange box) learns the drawing styles from a set of samples and generates drawings of a similar style for the input data.

and then visualized at each time step. To address this issue, GenerativeMap [55] proposes a generative model to synthesize a series of density map images and show the dynamic evolution between two given density maps, thus relieving the burden of storing intermediate results. Similarly, to visualize the dynamics in spatial-temporal data, TSR-TVD [67] applies a recurrent generative network to learn the pattern in dynamic evolution and generate temporal super-resolution visualizations. In these studies, ML learns the patterns in dynamic evolution from training data and directly synthesize intermediate visualizations to reduce the needed storage and computation cost. Another interesting direction is to facilitate parameter exploration in visualization creation. For example, Berger et al. [68] trained a Generative Adversarial Network (GAN) to synthesize visualizations and guide users in transfer function editing by quantifying expected changes in the visualization. Similarly, InsituNet [69] introduces a deep learning based surrogate model to create visualizations of simulation data by learning the mapping from simulation parameters to visualization images. These methods enable people to explore synthesized visualizations under various input parameters without actually executing the expensive creation processes.

4.3 Insight Communication

Process Description: In *Insight Communication* (Fig. 4(a)), insights are transformed into visualizations that can effectively communicate them. As with many studies in visualization community [70], [71], we refer the term *Insight* to the knowledge about the data that is communicated through visualizations. The main difference between *Insight Communication* and *Data Presentation* is the availability of insights. In *Data Presentation*, insights are hidden in the underlying data and need to be discovered by people through visual exploration. In *Insight Communication*, on the contrary, insights are already available to designers and need to be highlighted in the created visualizations. This difference is also denoted as exploratory versus explanatory visualization in prior studies [72]. To better communicate insights, visualizations in *Insight Communication* are usually more visually pleasing to improve user engagement and memorability.

Purposes of Employing ML: In *Insight Communication*, existing studies have successfully applied ML techniques to interpret the insight and to build a mapping from insights to suitable visual representations.

▸ *Interpreting insights.* The insights can be either provided by the designers or automatically generated from the data. For example, Text-to-Viz [56] allows users to provide insights through natural language statements such as *"40% of USA freshwater is for agriculture"* (Fig. 4(b)). A text analyzer (a supervised CNN+CRF model) is trained to extract entities from the sentence and understand the provided insights. DataShot [73] automatically identifies insights from tabular data and presents these identified insights to the users through infographics.

▸ *Mapping insights into visualizations.* ML techniques also demonstrate the capability to learn the mapping from insights to suitable visual representations. To facilitate this mapping, ML4VIS studies usually describe visual representations using specifications in a predefined design space to ensure a well-defined output space. Using the information from insights, ML can identify candidates for each dimension in the design space and produce the final visualization from these valid combinations [56], [57], [73]. For example, DeepEye [57] employs declarative visualization languages similar to Vega-Lite to enable the creation of common visualizations (*e.g.*, bar, pie charts). A decision tree model learns the mapping from insights and data characteristics to visualization specifications. Meanwhile, to generate visualizations with higher aesthetic values in *Insight Communication*, researchers also contribute more advanced design spaces. For example, Text-to-Viz [56] summarizes four design-space dimensions of infographics–layout, description, graphic, and color–from discussion with design experts. A visualization generator synthesizes infographics by generating a tuple of values on these four dimensions.

4.4 Style Imitation

Process Description: *Style Imitation* (Fig. 5(a)) refers to the process of generating visualizations that have similar styles with given examples. We use the term *Style* to denote a set of variables that affect the aesthetic appearance of visualizations without influencing how data are encoded. The *Style* of a visualization is related to a number of variables such as the color palette, the chart decoration, and the view layout.

Purposes of Employing ML: Even though the visualization style involves a large number of variables, existing ML4VIS studies on *Visualization Style Imitation* mainly apply ML techniques on two aspects, the imitation of color selection and the imitation of layout styles.

▸ *Imitating color selection.* Color selection in visualizations is not only an aesthetic choice but also influences the effectiveness of information communication. Even though some helpful guidelines about colors are available, it is still challenging for novice visualization developers to use these loosely-defined qualitative guidelines to generate high-quality color ramps. To address this issue, Color Crafter [58] mimics the practices of professional

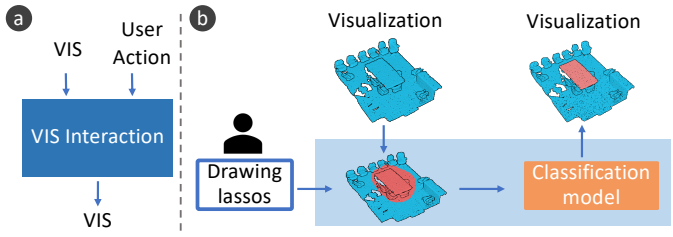

Fig. 6. (a) An illustration of the *VIS Interaction* Process. (b) A representative example of employing ML in *VIS Interaction*: LassoNet [59]. An MI model (orange box) predicts the user intended node selection in a 3D point cloud from a lasso drawn on 2D space.

designers and automatically generates high-quality color ramps. The color style imitation is achieved by modeling the paths that expert-designed color ramps traverse through color space. The authors trained ML models in a corpus of 222 expert-designed color ramps that are summarized from popular visualizations, and used these models to generate effective color ramps for developers from one single seed color.

▸ *Imitating layouts.* A parallel line of research on *Style Imitation* focuses on generating visualizations of similar layouts.

ML techniques have been extensively employed to generate graph drawings that mimic the layout styles of given graph drawing examples. Kwon et al. [74] presented an ML approach to facilitate large graph visualization by learning the topological similarity between large graphs. Specifically, the method provides a quick overview of visualizations for an input large graph data by learning from the drawings with similar topology structures. This approach is further extended in a recent study [75], which learns the layout from examples using a deep generative model. The model is then used to provide users with an intuitive way to explore the layout design space of the input data. Meanwhile, DeepDrawing [2] trains a graph-LSTM to learn one specific layout style from graph drawing examples. Instead of drawing a single graph in diverse layouts, the trained ML model in DeepDrawing directly maps new input data into graph drawings that share similar layout styles with the training examples, as shown in Fig. 5(b).

Apart from graph drawing, a few studies have explored imitating the layouts of other types of visualizations, including storyline visualization [76] and mobile visualization [77]. For example, PlotThread [76] trains a reinforcement learning agent to learn the layout of human-designed storyline visualizations. By imitating the layout from human-drawn examples, PlotThread is able to include more diverse narrative elements and create more expressive storylines compared with optimization-based methods.

4.5 VIS Interaction

Process Description: In *VIS Interaction* (Fig. 6(a)), the appearance of a visualization is modified based on the user actions (*e.g.*, zoom, filter). *VIS Interaction* usually reflects user intentions in visual exploration and modifies the visualization by manipulating the view configurations, the visual mappings, or the underlying data [78].

Purposes of Employing ML: In *VIS Interaction*, previous studies have demonstrated the capabilities of ML techniques to infer user intentions and thus assist users in interactions. According to our survey, the capabilities of the employed ML techniques in *VIS Interaction* can be classified into four main groups: refining interaction results, predicting next-step interactions, understanding natural interactions, and learning about users.

▸ *Refining interaction results.* ML techniques can refine the results of user interactions to achieve more accurate and efficient interactions. When interacting with a large number of visual elements, an accurate and efficient selection is crucial but usually hard to achieve. Take the brushing in 2D space as an example. Traditional interactions often need to make a trade-off between interaction efficiency and accuracy, *e.g.*, rectangular brushing is fast but inaccurate while the logical combination is accurate but slow. Fan and Hauser [79] exploited a CNN model to achieve both fast and accurate node selection in 2D scatterplots. The CNN estimates the intended node selection based on a simple line-brushing and data distribution in the visualization. Compared with selections in 2D space, node selection in 3D space is even more challenging: visualizing 3D points in 2D space easily causes occlusion; input devices such as touchscreens only operate in 2D space. To facilitate the node selection in the large-scale 3D point cloud, LassoNet [59] uses a deep learning model to predict the node selection based on a lasso drawn on a 2D surface, the user's viewpoint, and characteristics of the point cloud. A pipeline of LassoNet is illustrated in Fig. 6(b).

▸ *Predicting next-step interactions.* ML techniques can predict next-step interactions and facilitate complex visual explorations that consist of a series of interactions. Therefore, the employment of ML can ease the learning curve for users or reduce the interaction latency in complex visual explorations. For example, to ease the learning curve, REACT [80] models the analysis context using a generic tree-based model, in which the edges represent the user's actions and the nodes represent the system states. Given an analysis context, relevant contexts are first fetched using a context-similarity metric and then used to generate next-action suggestions for the user. To reduce the interaction latency, Ottley et al. [60] utilized a Markov model to predict the user's next mouse clicks in large scatterplots and fetch related data in advance.

▸ *Understanding natural interactions.* Natural interactions are interactions that humans naturally communicate through, such as gestures, natural language, and sketches [81]. ML techniques have demonstrated the capabilities to understand nature interactions and allow a more intuitive and convenient way to interact with visualization. Given that the input natural interaction (*e.g.*, sentence, sketch) can be fuzzy and uncertain, ML techniques help interpret the user's input and trigger dedicated functions. For example, ShapeSearch [61] allows users to search visual patterns from trend line visualizations using sketches and text descriptions. The authors trained a CRF model to recognize entities and understand natural interactions. FlowSense [82] supports users in visual data exploration via natural language command, *e.g.*, *"highlight the selected cars in a parallel coordinates plot"*. While the language understanding is mainly accomplished by a non-ML method (semantic parsing) in FlowSense, a supervised learning model is trained to effectively resolve syntactic ambiguity in the language commands.

▸ *Learning about users.* Interactions with the visualization can reflect users' reasoning processes in visual exploration and even their own personal characteristics. ML can effectively analyze these interactions and help designers learn about the users, including their learning abilities [83], their cognitive abilities [84], their analysis goals [85], [86], and even their personalities [62]. For example, Gramaz et al. [85] demonstrated that the classification of mouse interaction can effectively infer the tasks of cancer genomics experts. The ML classification can even expand current knowledge about the visualization tasks in cancer genomics. Brown et al. [62]

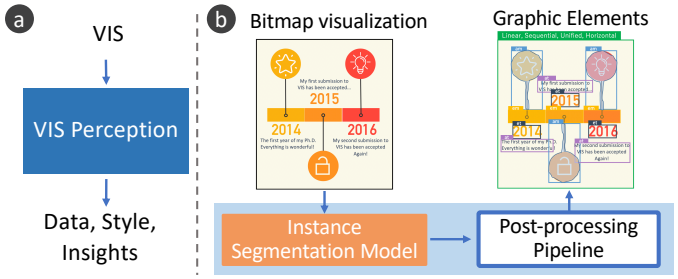

Fig. 7. (a) An illustration of the *VIS Perception* process. (b) A representative example of employing ML in *VIS Perception*. Chen et al. [5] employ a deep learning model (orange box) to extract the graphical elements from a timeline infographics.

applied ML techniques to analyze users' mouse interaction in the visual exploration of finding Waldo. They found that ML can accurately predict the user task performance (*i.e.*, the time used to find Waldo) and uncover the user personalities (*e.g.*, locus of control, extraversion, and neuroticism). Apart from mouse interactions, eye gaze data can also be treated as a type of interaction data and help learn about users [87]. Lallé et al [83], used ML techniques to predict a user's learning curve of a visualization based on the eye gaze data. By learning about users, visualization developers are able to go beyond the one-size-fit-all design and provide adaptive visualizations for different task needs, user abilities, and analysis stages.

4.6 VIS Perception

Process Description: In *VIS Perception* (Fig. 7(a)), users observe the appearance of a visualization, read the encoded data, and understand the underlying information. ML techniques have been used in *VIS Perception* process to automatically *"read"* the visualizations like humans, thus helping designers better understand users and enabling the analysis of a large corpus of visualizations.
Purposes of Employing ML: In *VIS Perception*, the capabilities of ML techniques can be divided into three main groups: extracting content, understanding content, and modeling users.
▸ *Extracting content.* ML techniques can directly extract content from visualizations in the form of bitmap images, such as graphical elements [5], color palette [88], and visual encodings [89]. The extracted content can be used for indexing, restyling, and reusing visualizations. A typical example is [89], which contributes an end-to-end pipeline to extract Vega visualization specifications from standard visualizations such as line charts. A set of classification models are trained to identify chart types, localize text, and recognize graphic marks separately. Compared with standard visualizations, infographics have more diverse appearances and are more difficult to interpret. Chen et al. [5] utilized a deep learning model, Mask RCNN, for instance segmentation to extract graphical elements from infographics. The extracted graphical elements are used as visualization templates for creating similar infographics using different data.

Meanwhile, since contents of bitmap visualizations are usually hard to be automatically extracted, some studies have proposed methods to embed the required information (*e.g.*, meta data, color schema) into bitmap visualizations [90], [91]. The embedded information will not influence human perception of the visualization and can then be easily extracted from the visualization. For example, VisCode [90] trains an encoder-decoder network, in which the encoder embeds a QR code to the background of a bitmap visualization and the decoder extracts the QR code from the visualization. A BASNet is trained to identify the semantically important regions and ensures the encoder will not affect human perception of these regions.
▸ *Interpreting content.* More than extracting content, ML techniques can also achieve a high level interpretation of visualizations and solve complex tasks. For example, Kembhavi et al. [92] proposed an LSTM-based method to understand the concepts and relationships in a diagram. Questions about the diagram like *"will the increases in lion lead to the decreases in deer?"* can be answered by an attention-based model with high accuracy. Another recent example DVQA is presented by Kafle et al. [63]. DVQA combines a CNN-based and an LSTM-based network and can answer questions about data visualizations. Given a visualization, the model reads the visualization content and answers reasoning questions such as *"which item sold the most units?"*
▸ *Modeling perception.* ML techniques can model how humans perceive a visualization and help designers better assess visualizations and understand audiences. A number of ML-based methods have been contributed to model user perception of a visualization from various aspects, including visual attention [64], similarity recognition [93], and readability [94]. For example, Bylinskii et al. [64] trained a deep learning model to predict the distribution of users' visual attention on an infographic. Guided by this prediction, designers can better arrange the layout and ensure that the most important contents are emphasized in the infographic. Haleem et al. [94] proposed a CNN-based approach to evaluate the readability of graph layouts by using graph images as inputs. An interesting phenomenon is that current ML4VIS studies extensively study the perception modeling in scatterplots, such as the perception of class separation [95], the perception of local patterns [96], and the perception of similarity [93]. This focus on scatterplots might stem from the wide popularity of scatterplots as well as their relatively simple visual forms.

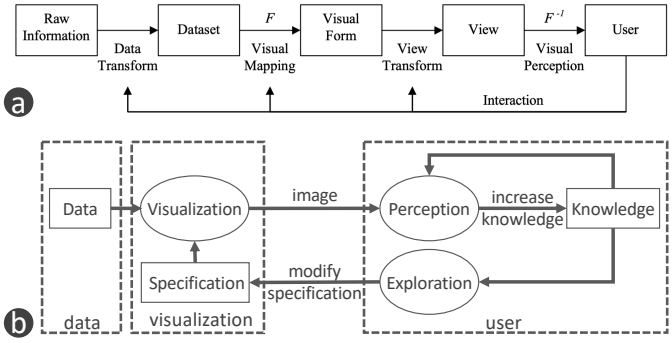

Fig. 8. (a) A visualization pipeline proposed by Card [40]. (b) A model of visualization proposed by Van Wijk [39].

5 An ML4VIS Pipeline

To map out the role of ML4VIS in general visualization and better inform the future exploration, we connect the six processes to previous visualization models and present an ML4VIS pipeline.

This ML4VIS pipeline is inspired by existing visualization models, especially the one proposed by Card [40] and the one by Van Wijk [39] (Fig. 8). We modify these visualization models to better fit the context of ML4VIS. Specifically, we make the following two main modifications. First, contrary to previous studies,

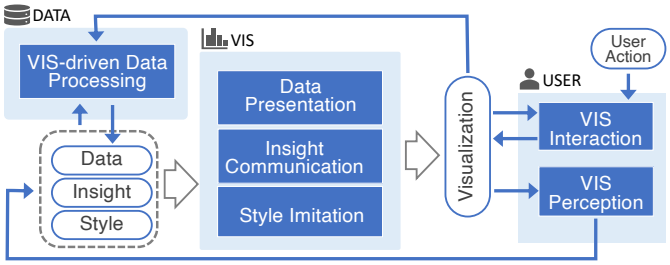

Fig. 9. An ML4VIS pipeline connects the six visualization processes that can be assisted by ML. Each blue rectangle presents a visualization process where ML can assist in; each white circle indicates a key element and can be either the input or the output of a process.

we take an additional element "style" into consideration. We refer to "style" as elements that affect the aesthetic appearance of a visualization, including color, layout, and chart decoration. "Style" is tightly related to the memorability [97] and engagement [98] of a visualization and is gaining increasing research attention. Second, we decouple this pipeline from users' internal states, *e.g.*, users' internal knowledge in Van Wijk's model, to better fit the context of ML4VIS. Instead of including users' internal states, we include the outputs of these internal states (*e.g.*, insights) to ensure elements in the pipeline are describable instance rather than vague concepts. For example, user actions can reflect users' intentions and tasks and insights can reflect users' perceived knowledge from a visualization. Since ML usually requires well-defined inputs and outputs, the modified pipeline can better fit in the context of ML4VIS by removing vague or hard-to-define concepts.

The ML4VIS pipeline is presented in Fig. 9. Each of the six blue boxes indicates a process where ML can assist in; each of the white boxes indicates a key element that is either the input or the output of a process. As with Van Wijk's model [39], the six processes are grouped by three containers: *DATA*, *VIS*, and *USER*.

This ML4VIS pipeline illustrates the close relationship among the six processes: the outputs of one process are used as the inputs for other processes. Therefore, employing ML techniques in one visualization process can further facilitate other processes, even though ML techniques are not used in these processes. An example is the study of Bylinskii et al. [64], which proposed a deep learning method to estimate users' visual attention on different regions of an infographic. While the employed ML techniques modeled user perception and contributed to the *VIS Perception* process, the estimated user attention can be later used in *Insight Communication* to help designers interactively modify the infographics layout and highlight important information.

6 Align ML with VIS

In this section, we review the current ML4VIS studies from an ML perspective, aiming to answer the question *"how is ML used to solve visualization problems"*. We categorize the collected ML4VIS papers based on the main ML tasks [50], [51] to provide an overview of how the needs in visualization are formed and solved as ML problems. Before the introduction of learning tasks, we also discuss the commonly-used formats of training data, which are summarized from the collected 85 ML4VIS studies using a bottom-up approach. The data formats, along with the learning tasks, decide how to choose an appropriate ML model to address the needs in visualization. We map the summarized data formats to the five types of information and the ML tasks to the six visualization processes. This ML-VIS mapping aims to provide both an understanding of the current practices and a guidance for future exploration on using ML in the ML4VIS research.

6.1 Training Data Formats

The format of training data reflects how information is encoded in ML. It can influence how to formalize ML problems and choose suitable ML models. The six visualization processes in Section 4 reveal five types of information: data (underlying the visualization), visualization, user action, style, and insight. In this subsection, we summarize the common data formats that are used to encode the five types of information based on a bottom-up analysis of the collected papers (as shown in Table 4).

Image is the format that most visualizations "in the wild" are presented in [108]. It is not surprising that many studies (*e.g.*, [88], [89], [99], [100]) encode **visualizations** in the form of images, contributing a number of large datasets of real-world visualization images: FigureSeer Dataset (60k) [102], AI2D dataset (5k) [92], Visually29K dataset (29k) [103], DVQA dataset (300k) [63], FigureQA dataset (100k) [101], ColorMapping dataset (1.6k) [88]. The emergence of large visualization-image datasets may result from the considerable differences between natural images and visualization images. Such differences make it difficult to directly apply the datasets and techniques developed for natural images to visualization images [5], [92], [107]. The **styles** of visualizations and the **insights** extracted from visualizations can also be encoded as parts of an image. For example, Chen et al. [5] encoded styles of an visualization as its segmented graphical icons. Bylinskii et al. [100] encoded the insights of an infographics as visual elements that are diagnostic of the topic of an infographics. Meanwhile, some researchers encode **user actions** in images. For example, Fan et al. [79] encoded the user actions (*i.e.*, click and

TABLE 4
Examples of Different Training Data Formats.

	Data	Visualization	User Action	Style	Insight
Image	—	[5], [55], [63], [69], [88], [89], [90], [99], [100], [101], [102]	[25], [79]	[5], [33]	[100], [103], [104]
Natural Language	—	—	[82], [105]	—	[56], [63], [99], [106], [107]
Engineered Features	[65], [66]	[108], [109]	[83], [84], [85]	[94]	—
Sequences	[54]	[54], [110]	[60], [62], [80], [111]	—	—
Graph	[2], [75]	—	—	—	[92], [107]

drag) in a scatter plot as the zoom and rotation of the scatter plot image.

Natural language (*e.g.*., text and speech) is the most common form of human communication [112]. Prior studies [82], [105] have enabled users to use natural language to express intended **user actions**, aiming to achieve more convenient and intuitive *VIS Interaction*. For example, Huang et al. [105] allowed users to interact with a visualization system using textual sentences such as "visualize trajectories passed through tourist attractions during January 25". Apart from user actions, **insights** can also be encoded in the form of natural language. Through natural language, users can easily describe the insights they aim to convey in *Insight Communication* [56] and understand the insights extracted in *VIS Perception* [63], [92], [106]. For example, Text-to-Viz [56] allows users to describe the insights using a natural language statement and automatically generates the corresponding infographics. AutoCaption [106] generates a text caption to describe the insights of a visualization based on four types of chart features detected (*i.e.*, aggregation, comparison, trend, distribution). In contrast to images, researchers usually collected a small number of labeled natural language examples for their specific tasks. For example, Yu and Silva [82] collected and labeled less than 20 examples to train an ML model that can resolve certain syntactic ambiguity; Cui et al. [56] collected and labeled 800 examples to train an ML model that can extract the four predefined entities. Also, due to the small number of training data, ML-based methods are often combined with non-ML-based methods. For example, Yu and Silva [82] supported natural language queries using non-ML-based semantic parsing. An ML model is only trained on a small number of labeled examples to resolve certain syntactic ambiguities.

Engineered features are high dimensional vectors that are extracted from the raw data using domain knowledge. While the great progress of deep learning helps reduce the reliance on human-centric feature engineering, extracting features using human knowledge is still popular in ML4VIS studies. This popularity of feature engineering may come from its ability to effectively capture important information without the needs of large datasets and enormous computational resources. Human specified features have been used to encode all the five types of information except insights. Here we discuss several representative examples. For the **data** (underlying a visualization), VizML [66] encodes a tabular data using 841 features related to statistical analysis, such as mean, standard deviation, entropy, and skewness. For the **visualization**, Beagle [109] encodes visualizations using the features extracted from SVG visualization files, such as number of axes, the position of circles, and the width of rectangles. For the **user actions**, Gramazio et al. [85] encoded user actions (*i.e.*, mouse clicks) through a set of human-defined features, such as the dwell time, the active time, the region-of-interest transitions. For the **style**, Haleem et al. [94] summarized the style of a graph layout using a set of graph aesthetic metrics, such as the node occlusion, the edge crossing, and the group overlap.

Sequences are rows of values whose order is important. Sequences have been widely used to encode **user actions** in their temporal order [60], [62], [111]. For example, Brown et al. [62] conducted sequence-based analysis of the user actions in visual analytics. The sequences of six types of user actions (*i.e.*, pan left, right, up, down, zoom in, and out) are converted into sub-sequences of two or three user actions, such as first zoom in and then zoom out. Interestingly, even though **data** (underlying visualizations) and **visualizations** do not have a sequential structure, they can also be encoded using sequences. For example, Data2Vis [54] encodes both JSON data file and Vega-lite visualization configuration as sequences. Table2Chart [110] encodes a visualization as a sequence of actions, including selecting data fields, selecting chart types, splitting data fields, and grouping data fields.

Graph is a non-Euclidean data format that consists of nodes and edges. **Data** (underlying visualizations) can be encoded in its original graph format in ML4VIS studies [2], [4]. Meanwhile, **insights** of visualizations can also be encoded in the form of graphs [92], [107]. For example, Kim et al. [107] encoded the insight of an infographic as a graph. Each node of the graph represents an object in the infographic and each edge represents the relationship between two objects.

6.2 Mapping ML Tasks to Visualization Processes

This subsection first provides a statistic summary of how the different ML tasks are mapped into different visualization processes. Then, we discuss each learning task in detail, including the related visualization processes, the commonly-used ML models, and the representative examples.

When summarizing the collected studies from an ML perspective, we find that it is not practical to summarize based on the employed ML models, as new ML models are constantly emerging and one ML model is often applied to solve different problems. To better align the capabilities of ML with the needs in visualization, we categorize the existing ML4VIS studies based on the types of ML tasks. This learning-task-based categorization allows us to understand how the needs in visualizations are formed and solved as ML problems, without diving into the ML model details (*e.g.*, whether to use LSTM or GRU).

Current ML methods can be categorized into four basic paradigms, *i.e.*, supervised learning, unsupervised learning, semi-supervised learning, and reinforcement learning [50], [51]. Based on the collected 85 papers, we refer to the learning tasks discussed in [50], [51] and further categorize the four paradigms into seven subsets: clustering, dimension reduction, generative, classification, regression, semi-supervised learning, and reinforcement learning. One subset of learning tasks is obtained when the following categorization is no longer related to learning tasks. For example, Nicolas [50] further categorizes reinforcement learning as Markovian or evolutionary. Since this categorization is about learning mechanisms rather than learning tasks, we treat reinforcement learning as one minimal subset of learning tasks.

6.2.1 An Overview

Fig. 10 shows the ML-VIS mapping between the six visualization processes and the ML tasks. Each column represents one type of learning task and each row represents one visualization process. In each cell, the number and the grayscale indicate the number of studies that are related to both the ML task in column and the visualizations process in row. The height of the bar indicates the number of studies related to a type of ML task (blue bar) or a visualization process (orange bar). Note that the number on a bar can be smaller than the sum of the numbers in corresponding cells, since a paper can involve more than one visualization process and one ML task.

Overall, most ML4VIS studies formulate the needs in visualization processes as supervised learning tasks (44 as classification and 25 as regression). Other types of learning tasks are less used.

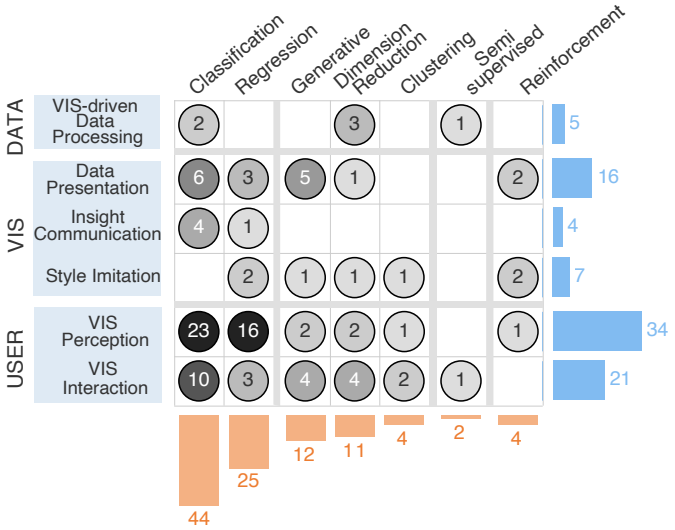

Fig. 10. Each row is a visualization process and each column is an ML task. In each cell, the number and the grayscale indicate the number of papers. The height of the bar indicates the number of studies related to a type of ML task (blue bar) or a visualization process (orange bar). The height of the bar can be smaller than the sum of the numbers in the corresponding row/column, because some papers involve multiple learning tasks/visualization processes.

Unsupervised learning, the second most-used learning, is employed in 27/85 ML4VIS studies. Only 4 papers use reinforcement learning and 2 papers use semi-supervised learning. This is not surprising as supervised learning is the most common type of learning [51]. When taking different visualization processes into consideration, the preference towards supervised learning, especially classification, appears on four of the six visualization processes. This preference is most pronounced in *VIS Perception*, where 23/34 studies are related to classification. Unlike other visualization processes, in *VIS-driven Data Processing* and *Style Imitation*, the collected ML4VIS studies do not exhibit a preference for a certain type of learning. But this observation needs to be treated cautiously due to the small number of studies in the two processes.

Among the six visualization processes, *VIS Perception* is the most frequently investigated visualization process in ML4VIS (by 34/85 papers). Most of the studies in *VIS Perception* employed supervised learning (23/34 used classification and 16/34 used regression). We speculate this is caused by the success of deep learning in computer vision tasks (*e.g.*, image classification, object detection, instance segmentation). A large number of deep learning models have been proposed to understand natural images and are adopted for the perception of visualization images [5], [31], [33], [64], [103]. A considerable proportion of ML4VIS studies investigate the problems in *VIS Interaction* (by 21 papers) and *Data Presentation* (by 16 papers). The other three processes (*i.e.*, *VIS-driven Data Processing*, *Insight Communication*, *Style Imitation*) are relatively less investigated.

6.2.2 Supervised Learning

In supervised learning, a model learns the mapping from input X to output Y from the labeled training examples. Labeled input-output pairs are required in supervised learning. Supervised learning can be further divided into **regression** and **classification** based on whether the output is numerical or categorical [51].

When formalizing a visualization problem as a **regression** task, the training examples need to be labeled with numerical values. Among all the six visualization processes, regression is mainly (19/25) used in the USER stage (*i.e.*, *VIS Interaction* and *VIS perception*). For example, in *VIS Interaction*, regression can be used to learn the dwell time of user mouse clicks [85], the users' learning curve of understanding a visualization [83]. In *VIS Perception*, regression can be used to learn the visually salient regions of a visualization [64], [103], the segmentation of important visual objects [5], and the aesthetic scores of visualizations [74], [94]. Regression is not used in DATA stage and less used in the VIS stage, which may be caused by the difficulty to quantitatively label the data processing and the visualization creation.

Many ML models can be used for regression tasks. How to choose a suitable ML model largely depends on the data formats. For example, CNNs are widely used when the input data are images. Many CNN models that are originally proposed for general computer vision tasks have been adopted to process visualization images, including ResNet (used in [102]), Xception (used in [104]), FCN used in ([48]), Fast R-CNN (used in [103]), and Mask R-CNN (used in [5], [31]). LSTM and GRU are used when processing sequences and natural languages [63], [107]. Traditional (*i.e.*, non-deep-learning-based) ML models (*e.g.*, SVR, logistic regression, MLP) are usually used when the inputs are engineered features.

When formalizing a visualization problem as a **classification** task, all the training examples need to be categorized into finite classes. Classification has been widely used in ML4VIS studies, such as evaluating whether a visual encoding is valid [65], predicting whether a visual region has interesting patterns [113], and recognizing the types of charts [99], [108]. Complicated problems can be decomposed into a series of classification tasks. For example, VizML [66] decomposes the creation of a visualization as a series of classification tasks, including classifying the type of marks, the type of shared axis, and the type of visualization. Kim et al. [107] and Kembhavi et al. [92] decompose the understanding of a diagram infographic as classifying the relationships between every two visual objects in the diagram. Classification has been used in all the six visualization processes except for *Style Imitation*. We conjecture this is because that the possible styles of a visualization can hardly be categorized into finite classes.

Most models used for regression can also be used for classification tasks through different ways, for example, setting a threshold (binary classification, 1-vs-the-rest classification) or using a softmax function (multi-class classification). Meanwhile, there are some ML models whose outputs are, by nature, discrete, such as SVM and decision tree.

The recent success of end-to-end deep learning methods have contributed many ML models that can accomplish regression and classification tasks at the same time. For example, Faster RCNN [114], an instance segmentation model, has been used in VIS perception and can predict the type (classification) and the bounding box (regression) of visual objects simultaneously [103].

6.2.3 Unsupervised Learning

In unsupervised learning, a model learns the underlying structure of the unlabeled data X. Compared with supervised learning, unsupervised learning relieves the requirement of the expensive data labeling process. Based on the collected papers, we discuss the three common tasks when using unsupervised learning in ML4VIS: generative learning, dimension reduction, and clustering.

Similar to prior research [4], this paper refers a **generative** model to a model that learns the distribution of unlabeled data and is capable of generating new samples that are similar to, but not the same as, the training data. Generative is mainly used in four visualization processes: *Data Presentation*, *Style Imitation*, *VIS Perception*, and *VIS Interaction*. In *Data Presentation* and *Style Imitation*, the creation of a visualization can be formed as generating a visualization that is similar to the collected visualization examples [55], [68], [69], [75]. In *VIS Interaction*, the prediction of user action can be formed as generating the next user actions, so that the action sequence is similar to the collected user action sequences [60], [80]. In *VIS Perception*, the human perception of a visualization can be modeled by learning the latent features of a visualization [115].

Among all the generative models, GAN is a hot research topic in the field of deep learning [116]. In ML4VIS, GANs have been exclusively used in *Data Presentation* for scientific visualization, including volume rendering [68], density maps [55], and in-situ visualization of ensemble simulations [69]. GANs can effectively synthesize visualizations without actually running the rendering process, which can be time-consuming and computationally-expensive. However, training GANs is usually difficult due to non-convergence and model collapse [117]. Therefore, traditional (non-deep-learning-based) generative models, such as Markov model, Gaussian mixture model, variational autoencoder, are also popular, especially when the training data is small and the problem is relatively simple. For example, a click event has a relatively simple data format compared with a visualization image and thus can be generated using a less complicated ML model. Ottley et al. [60] used a hidden Markov model to generate future click events on a scatter plot visualization. The evolution of user attention on the visualization is modeled based on the collected click stream data. The future click events are then generated based on the user attention evolution.

In **dimension reduction**, an ML model transforms the data from a high-dimensional space to a low-dimensional space. In the collected ML4VIS studies, dimension reduction is used in five visualization processes for two main purposes. In *VIS-driven Data Processing* and *Data Presentation*, dimension reduction is often used to process high-dimensional data for the visualization in a 2D space [53], [118]. For example, to visualize streaming data with varying dimensions, Fujiwara et al. [119] improved the incremental principal component analysis (PCA) and proposed a new dimension reduction solution. In *Style Imitation*, *VIS Perception*, and *VIS Interaction*, dimension reduction is often used to extract representative features for effective analysis [119], [120]. For example, to characterize the distribution patterns showed in scatter plots, Jo and Seo [120] trained a β-variational autoencoder to transform a scatter plot image ($64 \times 64 \times 1$) to a 32×1 feature vector. The 32 features capture the underlying data distribution of the scatter plot and can be used to predict the human-perceived distances between scatterplots.

Popular dimension reduction methods include PCA, MDS, t-SNE, UMAP, and autoencoder. PCA, MDS, t-SNE, and UMAP have been widely used for visualizing high dimensional data on the 2D space. While the liner methods such as PCA and MDS are good at preserving global structures, non-liner methods such as t-SNE and UMAP are good at preserving local structures. Autoencoders can be easily combined with other ML models to extract features from complicated inputs. For example, an autoencoder can be combined with graph neural networks (GNNs) to extract features from graphs [75], combined with 3D convolution to extract features from 3D stream lines and surfaces [121], and combined with CNNs to extract features from images [3], [120].

In **clustering**, a model divides unlabeled data into a number of groups based on their similarity. Clustering is a classical problem and has been used for many applications in the visualization field. In *Style Imitation*, collected training examples can be clustered based on the their color styles [58] to guide the creation of a new visualization. In *VIS Interaction* and *VIS Perception*, visual objects in a visualization can be clustered to better understand and interact with this visualization [115], [121], [122]. Generally speaking, clustering algorithms can achieve the most satisfactory results when the input features are representative. Therefore, when clustering complicate inputs (*e.g.*, images, natural language, extremely high-dimensional features), it is necessary to apply feature engineering or dimension reduction to extract representative features from the original data. We refer the readers to [51], [123] for more details about clustering algorithms.

6.2.4 Semi-supervised Learning

In semi-supervised learning. a model is trained using a small amount of labeled data with a large amount of unlabeled data. Semi-supervised learning is a combination between unsupervised learning (no data are labeled) and supervised learning (all data are labeled).

Overall, semi-supervised learning is a less explored type of learning in ML4VIS. The only two studies [52], [113] that employed semi-supervised learning belong to the same sub-category, *i.e.*, active learning. More specifically, Luo et al. [52] employed active learning to interactively clean data in *VIS-driven Data Processing*. Users are asked to label whether certain data items are duplicated and will influence the visualization. Their labels will then be used to improve the classification of data duplicates. Lekschas et al. [113] applied active learning to improve the interactive visual pattern search in *VIS Interaction*. A set of unlabeled visual regions are identified using an active learning strategy, and labeled by the users based on whether they contain interesting visual patterns. A classification model is then iteratively trained using the user labels to predict the interestingness of other unlabeled regions of the visualization.

6.2.5 Reinforcement Learning

In reinforcement learning, an agent learns to take actions in an environment to maximize the cumulative rewards. Different from supervised and unsupervised learning, the task of reinforcement learning is not to directly generate end results (*e.g.*, classification labels, latent features) but to take actions in an interactive environment (*e.g.*, playing a GO game).

To be solved through reinforcement learning, a problem needs to be formalized as finding a sequences of actions that maximizes the cumulative rewards. More specifically, the problem needs to be described as a Markov decision process, which contains a state space S, an action space A, a transition function, and a reward function. At each time step, the agent is in a certain state $s_t \in S$ and chooses an action $a_u \in A$. The agent moves to a new state s_{t+1} in the next time step based on the transition function $s_{t+1} = T(s_t, a_u)$ and gets a corresponding reward r based on the reward function $s_{t+1} = T(s_t, a_u)$ [124]. For example, PlotThread [76] formalizes the problem of creating a user-preferred storyline layout as performing a sequence of layout modifications on a less satisfied layout. All possible layouts form the state space and all possible

layout modifications form the action space. The reward is the similarity between the current layout and the user-preferred layout. The transition is implemented by applying the layout modification to the current layout.

Reinforcement learning has made significant success in a variety of tasks and a large number of reinforcement learning models have been proposed. However, only a limited number of ML4VIS studies have used reinforcement learning, including asynchronous advantage actor-critic [125] (used in PlotThread [76]), policy gradient [126] (used in MobileVisFixer [77]), and deep Q-learning [127] (used in Table2Chart [110]). How to apply reinforcement learning to ML4VIS still requires further exploration.

7 Research Challenges & Opportunities

In this section, we discuss the research challenges and opportunities of ML4VIS that are derived from our survey.

7.1 From the Aspect of ML

More Diverse Types of ML According to our survey, semi-supervised learning and reinforcement learning are less considered in ML4VIS studies, even though they demonstrate promising properties for ML4VIS studies.

Semi-supervised learning extracts knowledge from data that only a small proportion is labeled. By combining unlabeled data with labeled data, semi-supervised learning not only reduces the expense of data labeling but also surpasses both supervised learning (trained only using the labeled data) and unsupervised learning (trained using all data without their labels). For many problems in ML4VIS (*e.g.*, generating suitable visualizations), the data labeling requires human experts (*e.g.*, skilled designers). Sometimes, it is infeasible to get a large and fully labeled training dataset. The application of semi-supervised learning can help address this issue by reducing the required number of labels. Initial exploration of semi-supervised learning in ML4VIS have been conducted. For example, Lekschas et al. [113] and Luo et al. [52] have demonstrated the effectiveness of using active learning in data cleaning and visual pattern search. However, how to use more techniques in semi-supervised learning for more visualization processes in ML4VIS still requires future investigation.

Reinforcement learning is able to learn sequential decisions without enumerating all possible training examples. While current ML4VIS studies that use reinforcement learning are relatively sparse (4/85), previous ML studies have successfully applied reinforcement learning to human-computer interaction and demonstrated the capabilities of reinforcement learning in modeling user-data interactions [128], [129] and in making personalized recommendations [130]. These studies show the potential of reinforcement learning in various visualization problems, such as predicting next-step interaction in *VIS Interaction*, modeling users in *VIS Perception*, and recommending suitable visual representations in *Data Presentation* and *Insight Communication*.

Public High-quality Datasets In this survey, we notice that most papers need to construct their own datasets due to the lack of public visualization datasets. In existing ML4VIS studies, the dataset quality is often limited by the size of data and the reliability of label, and may potentially endanger the validity of the obtained ML models. For example, DeepEye [65] learns to classify "good"/"bad" visualizations based on the training examples labeled by 100 students, whose knowledge about visualization is unclear. VizML [66] trains a visualization recommender using collected online visualizations, yet previous studies have pointed out that online visualizations have a large proportion of deceptive visualizations [131]. These studies prove the effectiveness of ML in solving visualization problems, but the questionable quality of their training data can degrade the performances of the ML models (*e.g.*, the quality of the recommended visualizations). Public and high-quality datasets are needed to further improve the application of ML in visualizations.

Benchmark Tasks ML4VIS studies are still at their early stage and benchmark tasks for ML4VIS remain unclear. However, benchmark tasks are important for the progress of ML techniques [51]. In the field of ML, ML models are commonly evaluated by comparing their performances with the state-of-the-art on benchmark tasks (*e.g.*, question answering, instance segmentation). Advanced tasks can be accomplished by combining the methods developed for multiple benchmark tasks. The lack of benchmark tasks in ML4VIS makes peer comparison difficult and can lead to negative impacts on follow-up studies.

Meanwhile, we find that existing ML4VIS studies have intensively investigated several tasks, including graphic elements extraction [5], [89], visualizations generation [54], [65], [66], and graph layout [2], [74], [75]. These tasks can be the start point to define benchmark tasks in ML4VIS.

7.2 From the Aspect of VIS

Diversity in Visualization We notice that the forms of visualization are limited in the collected papers. First, current ML4VIS studies exclusively focus on static visualizations. Even though other forms of visualizations, such as animated transitions and data videos, are popular and effective in information communication, these visualizations are rarely discussed in the collected ML4VIS studies. We conjecture this is partly caused by the difficulty of understanding and generating dynamic visualizations using ML techniques. Take image-format training data as an example. Compared with statistic visualization (images), dynamic visualizations (videos) require a larger set of training data, more complicated ML models, and more powerful computational resources. Second, most ML4VIS studies focus on certain types of standard visualizations, especially bar charts, scatter plots, and graphs. Other types of visualizations, such as tree map, streamgraph, and parallel coordinates, are not discussed despite their wide popularity in real world. Moreover, chart decorations are overlooked even though they are important for the memorability and user engagement of visualization. Multi-view visualizations are rarely mentioned even though they are commonly used in visual analysis. Since the success of visualization depends on choosing proper forms of visualization [132], we believe good opportunities exist in applying ML techniques to more diverse visualizations.

7.3 From the Aspect of "4 (for)"

Towards VIS-Tailored ML In this survey, most ML4VIS studies directly apply ML techniques developed by ML researchers. However, general Ml techniques not always suit well for the specific problems in visualization. Take computer vision in *VIS Perception* as an example. General ML techniques are mainly developed for natural images and cannot be directly applied to images with charts and diagrams [5], [92]. Chen et al. [5] employed a deep learning model to understand timeline infographics, but

found that they needed a series of post-processing to address the poor model performance caused by the difference between natural images and charts. The authors also highlighted the importance to develop ML techniques specified for visualizations. We hope to see more studies that propose novel ML techniques to better solve the visualization problems.

Human-Machine Collaboration Most ML4VIS studies treat ML as a black box without the involvement of humans. Take *Data Presentation* as an example. ML directly generates suitable visualizations from the given data in an end-to-end manner [54], [66]. However, visualization is by nature a human-centric field: visual design often relies on the creativity of designers; visual analysis usually depends on the knowledge and experience of domain experts. Keeping human in the loop is crucial for the success of visualization. More importantly, a perfect ML model can rarely be obtained limited by the quality of training data and the ambiguity of the problem. Unsatisfactory model results will decrease user trust in ML4VIS and hinder the wide adoption of ML4VIS. Take *VIS Interaction* as an example, automated interaction refinement can confuse users when the prediction is at odds with the user's expectation [59], [79].

Considering the possibility of unsatisfactory model results, many ML4VIS studies allow user participation by supporting post-hoc refinement, such as adjusting the automatically selected nodes [59] and modifying the auto-generated visualizations [56]. Recently, initial exploration in human-machine collaboration has been conducted [76], [113]. For exampale, PlotThread [76] enables the ML agent and the human designer to work in the same canvas and modify the layout of a stroyline collaboratively. However, it requires further exploration how to support a close human-machine collaboration where human users are able to interpret, modify, and improve the ML. We envision that a close human-machine collaboration in ML4VIS can be achieved through explainable and interactive ML, which not only increases user trust by providing transparent predictions but only improves ML performances by utilizing user feedback.

Towards User-friendly ML4VIS The employment of ML not only provides opportunities but also poses new challenges in designing visualizations. For example, using ML to automatically refine user interaction or generate visualizations can violate the design principle of "minimize unexpected changes" [35], [133].

Some ML4VIS studies have discussed the usability issues of ML4VIS systems and proposed design suggestions in discussion and future work. For example, Wang et al. [2] suggested that self-exploring should be supported for expert users in automated graph drawing. When evaluating their proposed natural language interface for visual analysis, Yu et al. [82] found that users were often confused on why their queries were rejected and didn't know how to modify the queries. Yu et al. suggested that possible corrections of rejected queries should be provided to improve the usability of the system.

While these design suggestions for ML4VIS are insightful, they are scattered among different papers. A systematic summary of these design guidelines is still missing. More importantly, systematic cognitive studies are required to help designers better understand user behaviors and expectations in this new ML4VIS scenario.

8 Discussion

8.1 ML is Not a Panacea

In spite of the great success of ML4VIS, we still want to emphasize that ML is not the only technical solution towards intelligent visualization. ML-based methods are good at recognizing complicated patterns from a huge amount of data. However, when the pattern is simple or when training data is hard to obtain, ML may not be the optimal solution. Previous studies [134], [135] have demonstrated the effectiveness of non-ML-based methods, such as expert-defined rules and statistic summaries, in solving visualization problems, especially when the training data is hard to collect, when the problem space is sparse, or when the solution can be easily described through rules. For instance, ML techniques are widely used to understand visualizations in the form of bitmap images [5], [108]. But if the given input is an SVG file, a traverse of the SVG tree can effectively deconstruct the visualization and extract the content [136], [137], saving the great efforts to collect training data and the high computation cost to train an ML model. Another example is FlowSense [82], which uses a combination of pre-defined rules and ML methods to capture natural language input patterns due to the lack of labeled examples and computational resources. While we hope this survey can demonstrate the effectiveness of ML in visualizations and promote more ML4VIS studies, visualization researchers should also pay attention to the unique advantages of non-ML-based methods and apply them when applicable.

8.2 Limitations & Future Work

This survey comes with certain limitations due to the adopted approaches. First, we only investigate technique & application papers in this survey. Apart from technique & application papers, evaluation papers are also important for the field of ML4VIS. For example, Hearst et al. [138] conducted a crowdsourcing study with 274 participants to understand how to design natural language interfaces for visual analytics systems. A review of these evaluation papers can help us understand how users perceive ML4VIS applications and guide the design of ML4VIS. Second, we summarize the main purposes of employing ML in different visualization processes based on our analysis of the collected papers. We believe these goals can provide useful insights for future studies in the field of ML4VIS. However, it is possible that there are other under-explored areas of employing ML. ML4VIS is still an ongoing research field, we expect more studies to be conducted to expand our understanding about this field.

9 Conclusion

In this paper, we survey 85 papers to understand the current practices in ML4VIS research, *i.e.*, employing ML techniques for solving problems related to data visualization. Guided by two motivating questions: *"what visualization processes are assisted by ML?"* and *"what ML capabilities are used for visualization?"*, we summarize six main visualization processes that are benefiting from the application of ML techniques: *VIS-driven Data Processing, Data Presentation, Insight Communication, Style Imitation, VIS Interaction, VIS Perception*. For each visualization process, we summarize the main goals of employing ML with representative examples to illustrate the capabilities of ML techniques. To map out the role of ML4VIS in general visualizations, we referred to existing visualization models and organize the six visualization

processes in an ML4VIS pipeline. The six visualization processes are also aligned with the learning tasks in ML to reveal how the needs in visualization are formed as ML tasks. Current practices and future research opportunities in ML4VIS are discussed based on our analysis of current studies. We believe this survey can provide useful insights about the field of ML4VIS and promote future studies.

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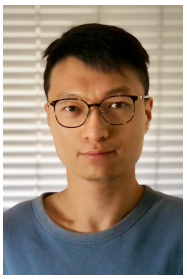

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